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PERFORMANCE EVALUATION IN TRAVEL ENGLISH FOR GA-ILMT

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ABSTRACT

We will describe the outline of the Machine Translation Method using Inductive Learning with Genetic Algorithms (GA-ILMT) and the experiments used in the performance evaluation for GA-ILMT, using translation examples of travel English. We performed the experiments using translation examples "on the plane" and "at the airport". These results show that GA-ILMT is adequately effective for the travel English.

KEY WORDS: machine translation, genetic algorithms, travel English, inductive learning

1 INTRODUCTION

Computer networks are rapidly becoming popular. As a result, society is quickly becoming international. However the language problem disturbs this internationalization of society. To resolve this problem, the machine translation system is very effective. The main method using machine translation systems is the Rule-based method [1] . The Rule-based method cannot deal adequately with various linguistic phenomena due to its use of limited rules. Therefore its correct translation rate is still low and the quality is poor. Moreover the improvement of the correct translation rate of the Rule-based machine translation system is difficult when this system has about two thousands translation rules. It is difficult because, even though the developer of the system adds one translation rule to the dictionary for correcting an error, this rule may cause several new errors. To resolve this problem, the Example-based machine translation [2, 3, 4] has recently been proposed. The correct translation rate and the quality of the Example-based machine translation system can be high because this method adapts to data automatically. However, this method requires many translation examples in its development.

To resolve this problem, Echizen-ya, Araki, and others previously proposed a method of Machine Translation using Inductive Learning with Genetic Algorithms [5] which has already been evaluated [6]. Echizen-ya, Araki, and others call this method GA-ILMT. Their research goal is to design a computer system with the same capability of language and knowledge acquisition as found in human beings [7, 8]. In GA-ILMT, translation rules are inductively acquired and translation results are produced by using these rules. Many new translation examples are automatically produced from only a small number of translation examples by applying genetic algorithms to the method of machine translation using inductive learning. The many new translation examples are produced by genetic algorithms do result in some errors. However, in the selection process of GA-ILMT, these errors are discovered from the translation results, and deleted from dictionaries. Echizen-ya, Araki, and others confirm that GA-ILMT is effective for solving this problem whereas the Example-based machine translation requires many translation examples [6]. However, in this experiment, the experimental data was taken from two first-year Japanese junior high school textbooks [9, 10]. Therefore the effectiveness of GA-ILMT for a conversational sentences is not still confirmed.

In this paper, we describe the experimental results of the performance evaluation of GA-ILMT for the travel English and some possible considerations. Also we propose that GA-ILMT is effective for the travel English based on the results of the experiments. GA-ILMT has learning capability. Therefore we consider that GA-ILMT can adapt to all kinds field data. In this experiment, we use the translation examples of English travel conversations from two different sentences which are "on the plane" and "at the airport". We confirm that GA-ILMT can adapt to two different sentences and show that GA-ILMT can be used for various purposes by its learning capability.

2 OUTLINE OF GA-ILMT

Figure 1 shows the outline of GA-ILMT. The experimental system based on GA-ILMT is the machine translation system from English to Japanese. First, a user inputs a source sentence in English. Second, in the translation process, the system produces several candidates of translation results using translation rules extracted in the learning process. Third, the user proofreads the translated sentences. Fourth, in the feedback process, the system determines the fitness value and performs the selection process of translation rules. In the learning process, new translation examples are automatically produced by crossover and mutation, and various translation rules are extracted from the translation examples by inductive learning. A translation example includes the source sentence and a translated sentence. There are two kinds of translation rules: those for sentences and those for words. The former are called sentence translation rules and the latter, word translation rules. Repetition of the above mentioned processes corresponds to generational replacement of the whole of system, and the system continuously evolves to higher quality translation.

As shown in Figure 2, a chromosome corresponds to a translation example which consists of an English and a Japanese sentence, and a gene corresponds to a word. In this paper, Japanese words are written in italics.

In the feedback process, the system evaluates the translation results using the translated sentences which have been proofread. The system determines the fitness value of the rules used in translation through the use of

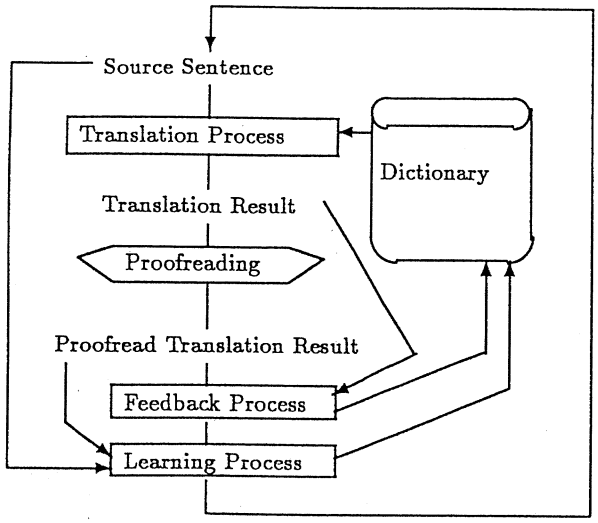


Figure 1: OUTLINE OF GA-ILMT

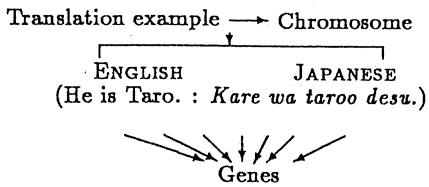


Figure 2: CHROMOSOME AND GENES

these correct and erroneous translation frequencies. The fitness value is calculated by the fitness function as follows:

$$\text{Fitness value}(\%) = \frac{\text{The correct translation frequency}}{\text{The number of uses}} \times 100 \quad (1)$$

And the system performs the selection process using the fitness value.

In the learning process, new translation examples are automatically produced by crossover and mutation. In crossover, two translation examples which have common parts are selected. Crossover positions are the common parts, and one-point crossovers for these two translation examples are performed. These one-point crossovers use each common part of the English and Japanese sentences. Figure 3 shows examples of crossover. The system extracts the common and different parts from the character strings of all translation examples. These common and different parts are used as translation rules.

In the translation process, the system produces several candidates of translation results for a source sentence using extracted translation rules. This process also uses genetic algorithm. The system substitutes the words in the word translation rules for the variables in the sentence translation rules. A translation rule includes a Japanese sentence or words corresponding to an English sentence or words. The system produces a Japanese sentence for the English sentence when the English sentence has the same character string as the source sentence. The Japanese sentence which is produced is the translation result. Figure 4 shows an example of how the translation result is produced. When there are several candidates of translation

(1)Selection from translation examples
ENGLISH JAPANESE
(He likes tennis.
:*Kare wa tenisu ga suki desu.*)
(She likes tea.
:*Kanojyo wa ocha ga suki desu.*)
(2)The crossover of the English sentence
He likes / tennis. → He likes tea.
She likes \ tea. → She likes tennis.
(3)The crossover of the Japanese sentence
Kare wa / *tenisu ga suki desu.*
Kanojyo wa \ *ocha ga suki desu.*
→ *Kare wa ocha ga suki desu.*
→ *Kanojyo wa tenisu ga suki desu.*
(4)The translation examples produced
ENGLISH JAPANESE
(He likes tea.
:*Kare wa ocha ga suki desu.*)
(She likes tennis.
:*Kanojyo wa tenisu ga suki desu.*)

Figure 3: CROSSOVER EXAMPLE

results, the system selects the correct translation result according to two criteria: the translation rule which has a higher fitness value and the translation rule which is more similar to the source sentence.

(1)The input sentence
I am your teammate.
(2)The initial group
ENGLISH JAPANESE
(I am @0. :*Watashi wa @0 desu.*)
(@0 am your @1. :*@0 wa kimi no @1 desu.*)
(@0 am @1. :*@0 wa @1 desu.*)
(3)The translation rule by crossover
ENGLISH JAPANESE
(I am your @1. :*Watashi wa kimi no @1 desu.*)
(4)The evaluation of population
ENGLISH JAPANESE
(teammate :*chiimumeito*)
(I am your @1. :*Watashi wa kimi no @1 desu.*)
(I am your teammate.
:*Watashi wa kimi no chiimumeito desu.*)
(5)The translation result
JAPANESE
Watashi wa kimi no chiimumeito desu.

Figure 4: EXAMPLE OF THE TRANSLATION PROCESS

3 EXPERIMENT

We evaluate the performance of GA-ILMT using translation examples of the English travel conversations "on the plain" and "at the airport" from English to Japanese. The conversations of two different scenes which are "on the plain" and "at the airport" are used in these experiments. We call "on the plane" and "at the airport" Scene 1 and Scene 2 respectively.

3.1 STANDARDS FOR EVALUATION

In these experiments, a correct translation result defines an exact same character string or a very similar expression as that of a translator. These evaluation standards are very

Table 1: NUMBERS OF DATA

No.	Scene 1	Scene 2
1	111	103
2	83	96
3	58	155
4	78	160
5	78	75
6	41	69
7	44	39
8	68	164
9	13	84
10	39	94
Total	613	1039

high. However translations like this are required by today's users.

3.2 DATA AND PROCEDURE

In these experiments, we used 613 translation examples of Scene 1 and 1039 translation examples of Scene 2. This data was taken from 10 English travel books for Japanese. These books are [11, 12, 13, 14, 15, 16, 17, 18, 19, 20]. We call these books No.1, No.2, ..., No.10 respectively. Table 1 shows the numbers of the data.

And we used two other machine translation systems which are developed by A and B companies for comparison with the system based on GA-ILMT. We call these two machine translation systems A, B respectively. We performed two experiments. One used data of Scene 1, the other used data of Scene 2. The initial dictionary of GA-ILMT was empty each experiment. The conditions of the selection process were that the number of uses would be over 5 and the fitness value would be under 25%. These thresholds were determined by a preliminary experiment. Translation examples were randomly changed by mutation, at a rate of 2%. The procedure of the experiments is Figure 1.

3.3 RESULTS AND DISCUSSION

Figure 5 shows the change of the correct translation rate for the data of Scene 1. Figure 6 shows the change of the correct translation rate for the data of Scene 2.

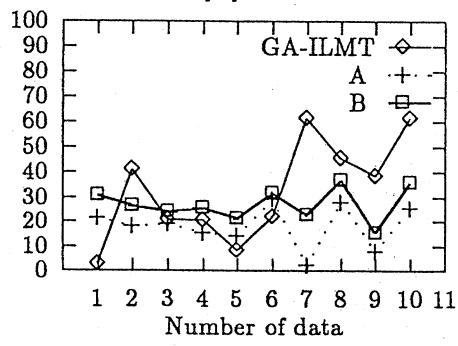


Figure 5: CHANGES OF THE CORRECT TRANSLATION RATE FOR THE DATA OF SCENE 1

The resulting data of Scene 1 experiment shows that the total correct translation rates of GA-ILMT, A and B are 27.4%, 19.1% and 28.1% respectively. However the

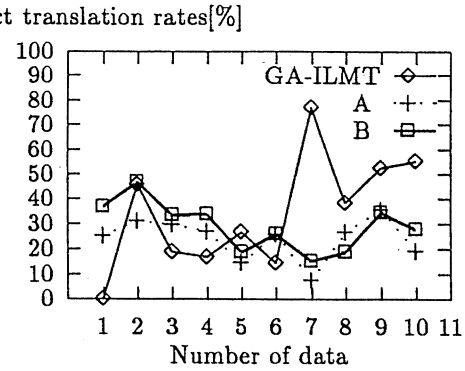


Figure 6: CHANGES OF THE CORRECT TRANSLATION RATE FOR THE DATA OF SCENE 2

total correct translation rates after the seventh book are 53.0%, 18.9% and 31.1% respectively. The resulting data of Scene 2 experiment shows that the total correct rates of GA-ILMT, A and B are 30.7%, 25.9% and 30.1%. However the total correct translation rates after the seventh book are 49.6%, 24.9% and 24.1% respectively. The total correct rate of GA-ILMT after the seventh book is the highest among these three systems. Therefore we confirm that the learning capability of GA-ILMT is high enough so that this system can adapt to Scene 1 and Scene 2.

Moreover, these experimental results show that the machine translation system which can perform a contextual translation has been developed by GA-ILMT. For example, in Scene 1, GA-ILMT translated "Beef, please" to "*Gyuuniku Ryouri wo Onegai simasu.*" (In Japanese "A dish of beef, please.") A translated it to "*Gyuuniku Douka.*" (In Japanese "Beef, please.") and B translated it to "*Gyuuniku wo Onegai simasu.*" (In Japanese "Beef, please.") In the context of Scene 1, "beef" means "a dish of beef". And GA-ILMT translated "Chicken, please" to "*Tori Ryouri ni shitekudasai.*" (In Japanese "A dish of beef, please.") A translated it to "*Douka, Okubyoumono dearu.*" (In Japanese "Be coward, please.") and B translated it to "*Chikin wo Onegai shimasu.*" (In Japanese "Chicken, please.") In the context of Scene 1, "chicken" means "a dish of chicken". And GA-ILMT translated "I feel sick." to "*Kibun ga Warui nodesuga.*" (In Japanese "I get airsick.") A translated it to "*Watashi wa Byouki noyouni Kanji rareru.*" (In Japanese "I feel sick.") and B translated it to "*Watashi wa Byouki noyouni Kanji masu.*" (In Japanese "I feel sick.") In the context of Scene 1, "sick" means "airsick". Thus the data shows that the translation results of GA-ILMT are correct and natural because these translation results fit the context of Scene 1.

In the experiment of Scene 2, GA-ILMT translated "What was your flight number?" to "*Nan Bin desuka?*". A translated it to "*Anata no Furaito Nunbah wa, Nan de attaka?*" and B translated it to"*Anata no Furaito Suu wa Nan deshitaka?*". These three translations all render the same meaning of "What was your flight number?". However "*Nan Bin desuka?*" rendered by GA-ILMT translation is more natural than the translation results of A and B in the context of Scene 2. And GA-ILMT translated "May I have a receipt?" to "*Keisansyo wo kudasai.*" (In Japanese "Please, give me a receipt.") A translated it to "*Watashi niwa Reshihto ga Arukamo shirenaika?*" and B translated it to "*Watashi wa Reshihto wo Motteiru kamo shiremasenka?*". In the translation results of A and B,

the "have" means "possess". Therefore they are wrong. And GA-ILMT translated "I have 2 bottles of whisky." to "*Uisukih wo 2 hon Motteimasu.*" (In Japanese "I possess 2 bottles of whisky.") A translated it to "*Watashi wa Uisukih no 2 bin wo nomu.*" (In Japanese "I drink 2 bottles of whisky.") and B translated it to "*Watashi wa 2 bin no Uisukih wo nomimasu.*" (In Japanese "I drink 2 bottles of whisky.") Generally, "have" means "drink" when the object for word "have" is "whisky". Both of the contexts used by A and B are the context found inside a sentence. However, "have" means "possess" in the context of Scene 2. The translation result of GA-ILMT is correct because GA-ILMT uses the context of the whole scene.

These results show that GA-ILMT is adaptable enough for these kinds of examples from English travel conversations. Therefore GA-ILMT records a higher-correct translation rate for English travel conversations than A and B after the seventh book. Furthermore the translation expressions of GA-ILMT are more natural than those of A and B. These results show that GA-ILMT can learn the context of Scene 1 and Scene 2.

4 CONCLUSION

We confirmed that GA-ILMT adapted to these two scenes and was adequate enough to obtain highly-correct and highly-qualitative translation results in these two scenes. The results of the experiments show that GA-ILMT has adequately reached a high performance translation level for these two limited scenes. However, in these experiments, the initial dictionaries were empty and GA-ILMT could only learn the context from translation examples through its own learning capability. Therefore we consider that GA-ILMT has high performance for various data of all fields. In the future, we plan to experiment using the various data of many other fields to confirm it.

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